

WaveKitchen

A Signal Processing Cooking Game

FULL-STACK SYSTEM ARCHITECTURE SPECIFICATION · v2.1

React 18 + Vite · FastAPI (Python 3) · SQLAlchemy 2.0 · SQLite / PostgreSQL
Server-Side Authoritative DSP (NumPy / SciPy) · 49 Automated Tests
Phased Array Acoustic Beamforming · Block-Based Signal Pipelines

1. Architectural Principles & Tier Model

WaveKitchen employs a three-tier architecture with a strict server-authoritative trust boundary. The presentation tier (React 18 SPA) renders waveforms, plots, and interactive controls; the application tier (FastAPI) executes all mathematical transforms, impulse response convolutions, phased array beamforming, and scoring; the data tier (SQLite / PostgreSQL) stores player state, recipes, and sessions.

Field	Specification Detail
Architecture Pattern	Three-tier client-server with authoritative backend DSP engine
Frontend Stack	React 18, Vite 5, TanStack React Router, Tailwind CSS, Canvas 2D, Web Audio API
Backend Stack	Python 3.13, FastAPI, Pydantic v2, SQLAlchemy 2.0, NumPy, SciPy
Database	SQLite (zero-configuration local file) / PostgreSQL via WK_DATABASE_URL
Transport Format	JSON over HTTP; audio signals encoded as base64 little-endian float32 arrays
Sample Rate & Frame	SR = 22,050 Hz canonical sample rate; FRAME = 4096 samples (~186 ms block length)
Automated Verification	49 automated tests (19 mathematical DSP assertions, 30 API integration tests)

2. Why DSP Lives on the Server

- 1. **Security & Integrity:** Clients never send audio signals or scores; they send only parameter and filter specifications.
- 2. **Reproducibility:** A session stores a random seed. The server regenerates identical contaminated waveforms on demand.
- 3. **Numerical Fidelity:** Double-precision 64-bit accumulations and optimized SciPy FFT / convolution routines prevent browser floating-point drift.
- 4. **Singular Reference Truth:** The reference dish and player dish are evaluated through the exact same Python function.

3. Backend Module Inventory

Module Path	Core Responsibility	Key Exports & Functions
app/dsp/core.py	Fundamental DSP math primitives	fft, ifft, stft_magnitude, lowpass, notch, convolve, time_scale, time_shift, am_modulate, decimate, encode
app/dsp/instruments.py	Acoustic harmonic synthesizers (18 ingredients)	synth, VOICES (marimba, drum, flute, shaker, kalimba, triangle, clarinet, organ, bassoon, glockenspiel, harp, oboe, piccolo)
app/dsp/contamination.py	Seeded stochastic noise generators	corrupt, Contaminant (white, hum, hiss, burst, tone, pink)
app/dsp/systems.py	LTI appliance impulse responses	ir, cascade_ir, ir_comb, ir_decay, ir_resonator (grill, fry, bake, boil, simmer, sear, smoke, steam)
app/dsp/beamforming.py	Phased array acoustic beam delivery	calculate_beam_angle, array_factor, generate_beam_pattern, check_beam_alignment, get_preset_phases_for_angle
app/dsp/pipeline.py	Canonical 6-stage transformation pipeline	run_pipeline, apply_filter_chain, reference_params, default_bands
app/dsp/metrics.py	Objective similarity & quality metrics	dish_metrics, prep_quality, stars, snr_db, spectral_cosine_similarity
app/gameplay.py	Gameplay service layer & dish judging	clean_signal, dirty_signal, compute_stages, judge, award, maybe_unlock
app/models.py	SQLAlchemy ORM database models	Player, Recipe, GameSession, SessionIngredient, Attempt, Ingredient, Appliance
app/schemas.py	Pydantic v2 API request/response contracts	SessionOut, SubmitResult, FilterResponse, BeamDeliveryRequest, BeamDeliveryResponse
app/routers/sessions.py	Session lifecycle & station endpoints	start_session, apply_filters, accept_ingredient, set_params, convolution, submit, beam_delivery

4. REST API Endpoint Specification

Method & Path	Purpose	Auth
GET /api/health	System health, sample rate (22050 Hz), frame size (4096)	None
POST /api/players	Register chef handle, receive bearer token	None
GET /api/players/me	Profile, career points, tier rank, best scores	Token
GET /api/recipes	Recipe book catalogue (targets withheld for gameplay)	Optional
GET /api/recipes/{id}	Full recipe details including target values	None
GET /api/ingredients	Catalogue of 18 ingredients with spectral signatures	None
GET /api/appliances/{id}	Impulse response array $h[n]$ and $ H(f) $ frequency curve	None
POST /api/sessions	Start new service, generate deterministic seed, contaminate produce	Token
POST /api/sessions/{id}/ingredients/{s}/filter	Apply graphic EQ + filter tools, return reconstructed waveform & FFT	Owner
POST /api/sessions/{id}/ingredients/{s}/accept	Cleanliness counter check (409 if still dirty or over-filtered)	Owner
PUT /api/sessions/{id}/params	Update cooking parameters, re-run pipeline, return all stages	Owner
GET /api/sessions/{id}/convolution?lag=m	Interactive sliding convolution partial sum at lag m	Owner
POST /api/sessions/{id}/submit	Server-side judging, 4-part score breakdown, star awards, tier unlock	Owner
POST /api/sessions/{id}/beam-delivery	Phased array acoustic beam delivery steering & alignment check	Owner
GET /api/leaderboard	Global & per-recipe leaderboards and rankings	None

5. Automated Verification & Quality Assurance

The test suite contains **49 automated tests** that assert exact mathematical identities and full API flows:

- **FFT Lossless Inversion:** $\max |x - \text{IFFT}(\text{FFT}(x))| < 1\text{e-}5$.
- **LTI Cascading:** Direct vs FFT convolution agreement ($< 1\text{e-}4$) and system associativity.
- **Time Scaling & Delay:** Inverted duration changes and linear phase preservation.
- **Phased Array Beamforming:** Array factor magnitude normalization and steering angle alignment within $\pm 6^\circ$.
- **End-to-End API Suite:** All 18 ingredients, 5 master recipes, dirty/clean delivery separation, filtering acceptance thresholds, and score breakdowns.